

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY 2021 TRIMESTER

FINAL ASSESSMENT

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

WEDNESDAY, 15 SEPTEMBER 2021

TIME : 2.00 PM – 5.00 PM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instructions to Candidates:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an Individual, time restricted assessment which consists of **FOUR (4)** questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **5.00 pm, 15 SEPTEMBER 2021**.
3. During the period of **3 hours** of this FA, the examiner(s) can be reached at
 - (a). Teams.microsoft.com (Please join 15 minutes earlier) code: **sm23sak** or
 - (b).Email: noraziraj@utar.edu.myYou may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.
4. You may refer to any books, lecture notes, published materials, online resources etc. when answering the questions. Proper referencing is necessary to avoid plagiarism. The preferred referencing style is American Psychological Association (APA) / Harvard / Chicago Manual. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Answer Script File**

5. The answer script **MUST** be a **Microsoft Word file**, in A4 size format. **Note: Please keep the file size NOT exceeding 30MB per file.**
6. Please check you Index Number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script file using the following file name for submission:

UCCD1203_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme IB, and your Index Number is A01234CBIBF, then your answer script should be named as

UCCD1203_FA_IB_A01234CBIBF.doc

Answer Script File Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.

(a). Attach your answer script at Google Form

(link: <https://forms.gle/CMpptiffpQHg6byGA>)

Note: Use your “1UTAR” email account to access the Google Form. You can edit your submission before the form is closed.

(b). Send your answer script to the following Email Account according to your programme:

i. CN and CT students please send your answer script to:

[UCCD1203FACNCT@gmail.com]

ii. CS students please send your answer script to:

[UCCD1203FACS@gmail.com]

iii. IA students please send your answer script to:

[UCCD1203FAIA@gmail.com]

iv. IB students please send your answer script to:

[UCCD1203FAIB@gmail.com]

Note: For the subject/title of your email, please use the file name of your answer script, i.e.

UCCD1203_FA_[Programme Abbreviation]_[Your Index Number]

8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). When multiple answer script files are received, the examiner will randomly choose one for marking.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Contents of Answer Script**

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page. It is recommended that the answer to each question is limited to **2 pages**.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size **12**, with proper spacing and alignment, except for the drawings and equations/calculations.
13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document, as part of your submitted answers.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

- Q1. A database consists of the list of travel agencies that offer various tour packages to different countries around the world as in the tables below. The customers may book the packages from any interested travel agencies and may pay for the packages in a series of installments and a receipt is issued for each payment. When the customer books a tour package, the agency is required to records the booking date, the total amount owed by customer and their current outstanding balance on the tour. The table below contain partial relations and records from travel agencies database.

AgentCode	AgentName	AgentCity	AgentPostcode	AgentContact	AgentEmail
AS70	Star Tour	Gombak	53100	03-8593018	startour@gmail.com
AM71	Majestic Holiday	Georgetown	10200	04-6891182	majestic@outlook.com
AB72	Bayu Travel & Tour	Kota Kinabalu	88000	088- 429888	bayuttravel@gmail.com
AW73	BlueWave Holiday	Kuching	93000	082-206021	admin.bluewave@gmail.com
AE74	EastWest Vacation	Seremban	70400	06-9184093	eastwest@hotmail.com
AK75	Kingdom Tour	Kuala Lumpur	53000	03-9492899	kingdomtour@outlook.com

Table 1.1 Partial contents of the Agency relation

CustNo	FName	LName	CustCity	Postcode	Contact	Email
C001	Dahlia	Razak	Kota Kinabalu	88000	012-9388194	dahlia99@gmail.com
C002	Priya	Kaur	Klang	41000	011-3819372	priyakaur@gmail.com
C003	Kamarul	Azlan	Seremban	70400	010-1836825	Kamarul123@hotmail.com
C004	Sarah	Lee	Kuching	93000	019-3864951	sarahlee@gmail.com
C005	Qisya	Azman	Georgetown	10200	012-3859274	qisyaa88@yahoo.com
C006	Prakash	Kumar	Alor Setar	05050	018-2123889	prakash32x@gmail.com
C007	Jie Lin	Chan	Kota Kinabalu	88000	010-1284999	jielin@yahoo.com
C008	Jesslyn	Tan	Kudat	89050	011-3839283	jesslynn@hotmail.com

Table 1.2 Partial contents of the Customer relation

TNo	TName	TMaxPart	DepartDate	ReturnDate	ChildCost	AdultCost	AgentCode
TS1	New Zealand Exploration	10	01/06/2019	12/06/2019	RM7700	RM9000	AS70
TS2	Japan Cherry Blossom Tour	20	01/03/2019	07/03/2019	RM2500	RM3000	AS70
TM2	Australia Discovery	20	15/11/2019	22/11/2019	RM1500	RM1800	AM71
TB1	Turki Adventure	30	31/12/2019	08/01/2020	RM1800	RM2300	AB72
TB3	Spectacular Morocco	20	28/02/2019	08/03/2019	RM2200	RM2500	AB72
TW1	New Zealand Flex Tour	20	01/12/2019	13/12/2019	RM8000	RM9000	AB73
TW3	Australia Golden Tour	25	21/05/2019	26/05/2019	RM1300	RM1500	AB73
TE1	Discovery Australia	13	14/12/2019	21/12/2019	RM1400	RM1700	AE74
TE3	Japan Escapade	10	18/05/2019	24/05/2019	RM2200	RM2500	AE74
TK3	Morocco Private Tour	20	24/02/2019	04/03/2019	RM2400	RM2700	AK75

Table 1.3 Partial contents of the Tour relation

BID	CustNo	TNo	BookDate	BookNoChild	BookNoAdult	Total	BookCost	BookOutstanding
BK101	C001	TS1	25/05/2018	1	2	RM19700	RM 5000	RM 14700
BK202	C002	TS2	14/12/2018	3	5	RM 22500	RM 7000	RM 15500
BK505	C005	TM2	13/03/2018	1	3	RM 6900	RM 1000	RM 5900
BK707	C001	TB1	16/10/2018	2	4	RM 12800	RM 3000	RM 9800
BK909	C008	TB3	07/08/2018	1	2	RM 7200	RM 500	RM 6700
BK111	C002	TW1	25/12/2018	1	4	RM 44000	RM 10000	RM 34000
BK333	C007	TW3	27/07/2018	2	6	RM 11400	RM 2500	RM 8900
BK444	C004	TE1	29/06/2018	2	5	RM 11300	RM 2500	RM 8800
BK666	C003	TE3	03/01/2019	1	3	RM 9700	RM 1000	RM 8700
BK999	C006	TK3	14/11/2018	2	6	RM 21000	RM 5000	RM 16000

Table 1.4 Partial contents of the Booking relation

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q1. (Continued)**

PayReceiptNo	CustNo	TNo	PayDate	PayAmount
RX010	C001	TS1	25/05/2018	5000
RX011	C005	TM2	13/03/2018	100
RX012	C008	TB3	07/08/2018	500
RX013	C007	TW3	27/07/2018	2500
RX014	C001	TS1	11/09/2018	2000
RX015	C004	TE1	29/06/2018	2500
RX016	C008	TB3	18/12/2018	300

Table 1.5 Partial contents of the Payment relation

TNo	HotelID	StopDate	NoOfDays
TS1	NZ01	02/06/2019	6
TS2	JP01	03/03/2019	3
TM2	AS02	16/11/2019	4
TB1	TK01	02/01/2020	5
TB3	MR01	02/03/2019	3
TW1	NZ02	04/12/2019	6
TW3	AS01	21/05/2019	3
TE1	AS03	17/12/2019	2
TE3	JP02	18/05/2019	3
TK3	MR01	28/01/2019	4

Table 1.6 Partial contents of the Stop_Area relation

HotelID	HotelName	HotelCity	PostCode	HotelEmail
NZ01	NZ Queen	Queenstown	9300	nzqueen@gmail.com
NZ02	The Garden	Christchurch	7476	thegarden@gmail.com
JP01	The Lake	Osaka	530-0000	thelake@gmail.com
JP02	TwinSwan	Tokyo	100-0001	twinswan@gmail.com
AS01	SmallTown	Sydney	2000	smalltown@gmail.com
AS02	PrimeLand	Perth	6000	primeland@gmail.com
AS03	BreezeView	Melbourne	32901	Breezeview@gmail.com
TK01	King Inn	Cappadocia	67060	kinginn@gmail.com
MR01	Voyage Cottage	Casablanca	20070	voyagecot@gmail.com

Table 1.7 Partial contents of the Hotel relation

- Write **ONE (1)** SQL (Structured Query Language) statement to calculate the total booking trip to 'Morocco' (refer to TName) where the departure date was happened during the first half of the year 2019. (5 marks)
- Write **ONE (1)** SQL statement to list all the available tour in December but having maximum participants less than 25 people only. Arrange the list based on the alphabetical order of the tour name. (4 marks)
- Write **ONE (1)** SQL statement to list all the customer's name with their selected travel agency who had booked the trip to 'Tokyo' or 'Australia', however still having outstanding payment with less than RM15000. (7 marks)
- Write **ONE (1)** SQL statement to list the agencies that offer a stay trip to 'Cappadocia' with the minimum overall travel period of 7 days. (6 marks)
- List the table creation sequence for the travel agencies database. (3 marks)

[Total : 25 marks]

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

- Q2. (a) The ERD in Figure 2.1 shows the product purchasing by the customers from the manufacturer.

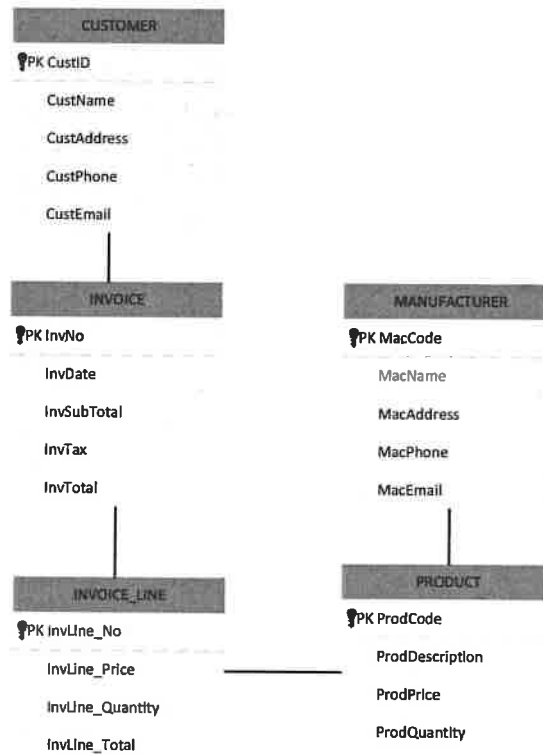


Figure 2.1 The partial entity relationship diagram (ERD) of product selling records

- (i) Identify and explain **ONE (1)** time dependent component. (3 marks)
 - (ii) Identify **TWO (2)** derived attributes and specify how the value is derived. (4 marks)
 - (iii) Identify **THREE (3)** foreign key(s) in Figure 2.1. Indicate which table and attribute the foreign key references. (3 marks)
- (b) The relational schema shown below is part of EasyLogistic Management company's database. Note that one branch is managed by a manager, each employee is allocated at one branch only and each employee will be assigned under one supervisor except the manager.

Employee (empID, LastName, FirstName, position, salary, branchNo, supervisorNo)

Branch (branchNo, street, city, state, zipCode, phoneNum, managerID)

- (i) Identify the foreign key for each of the table by including the table and attribute the foreign key references. If a relation does not have a foreign key, write NONE. (3 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q2. (b) (Continued)**

- (ii) The manager of the branch will change over time. Modify the relational schema above to maintain the record of the branch manager. (5 marks)

- (c) The following relational schema form part of a database held for library borrowing system. BookCopy is referring to the physical copies of each book in the library.

Book (BookID, BookTitle, BookAuthor, publishDate, publisherName, ISBN)

BookCopy (copyNo, ISBN, Price)

Borrower (BorrowID, BorrowName, BorrowPhone, BorrowEmail)

Loan (BorrowID, dateTimeOut, dateDue, dateReturned, copyNo)

- (i) Identify **ONE (1)** weak entity based on the above relational schema. Justify your answer. (3 marks)
- (ii) Explain how the entity integrity and referential integrity rules are applied to these relations. (4 marks)
- [Total : 25 marks]

- Q3. (a) An agency called FindStaff provides part-time/temporary staff to hotels throughout Northern Region of Malaysia. The table shown in Figure 3.1 lists the working details of the agency staff.

ContractNo	StaffID	hoursPerWeek	staffName	hotelID	hotelLocation
Jan2020	SC19482	18	Adam Malik	H03	Langkawi
Jan2020	SL18117	12	Sarah Ahmad	H03	Langkawi
Mac2020	SC19482	24	Adam Malik	H15	Taiping
Mac2020	SM18003	20	Daniel Kim	H15	Taiping
May2020	SG20129	28	Catherine	H03	Langkawi
May2020	SL18117	24	Sarah Ahmad	H15	Taiping

Figure 3.1 Staff and their contracts information to work at hotels

- (i) Refer to Figure 3.1, which of the attribute(s) form a primary key for this table? Justify your answer. (3 marks)
- (ii) The table shown in Figure 3.1 is susceptible to update anomalies. Provide **ONE (1)** example of insertion, deletion and modification anomalies. (6 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q3. (Continued)**

- (b) Figure 3.2 shows a flight booking confirmation issued by FlightCatches website.

Booking No	R1101	R1101	R1206	R2682
Booking Date	24/06/2020	24/06/2020	07/06/2020	12/04/2020
Visa Card No	5283-2912-412	5283-2912-412	1281-9421-385	6342-1246-341
Ticket Price	RM3000	RM3700	RM3200	RM400
Customer ID	C3700	C3700	C4999	C1220
Customer Name	James Colly	James Colly	Anne Tinker	Sarah Adam
Customer Address	No.12, Quebec	No.12, Quebec	No.85, Bandar Seri Begawan	No.14, Selayang
Flight Number	AC850	BI98	BI98	MH0783
Flight Date	28/07/2020	04/08/2020	04/08/2020	14/05/2020
Departure Country	Canada	Brunei	Brunei	Malaysia
Departure Time	1400	0500	0500	1700
Arrival Country	Brunei	Canada	Canada	Indonesia
Arrival Time	0600	2100	2100	1900

Figure 3.2 The Flight Booking Details

- (i) Draw the dependency diagram showing all the dependencies in the relation, based on the details shown in Figure 3.2. (8 marks)
- (ii) Illustrate the process of normalization by converting the table from the First Normal Form (1NF) to Third Normal Form (3NF). (8 marks)
- [Total : 25 marks]

- Q4. (a) Connection traps is a problem which often occur due to a misinterpretation of the meaning for certain relationships between entities.
- (i) Given three entities named Faculty, Department and Student. Draw a simple ERD to illustrate the fan trap situation. (2 marks)
- (ii) Reconstruct the answer in Q4(a)(i) to eliminate the fan trap. (2 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q4. (Continued)**

- (b) As a database consultant, you have been asked to develop an ERD for the event management company named Majestic Event Planner. The information stored on the booking event include the event name, location, capacity, type, start and end date of the event. For each event, there will be performers assigned to it. The data stored on performers includes the performer id, name, home address, phone number, email address and genre performance. The performer id uniquely identifies each of them. Each performer can perform at various event during different period of times. The company also provides the equipment to be rented during the event. The equipment is categorized based on the type such as lighting equipment, stage construction, musical instruments etc. Each type of equipment is differentiated using unique values. Besides, it is also important to store the detailed information of the equipment for each type such as equipment name and their quantities. In most cases, multiple equipment will be rented during the event and are reusable for another event. Therefore, it is necessary to keep track the equipment usage such as the quantity of each rented items and the cost for each of it. A few staffs will be assigned to handle each event at one time. Depending on time, they will be assigned to another event when available. Information is stored on the staff including staff number, name (first and last name), home address, home phone number, date of birth (DOB), position and salary.
- (i) Draw an ERD from the above scenario that includes all the entities, attributes of each entity, relationships, and the multiplicity of the relationship, i.e., cardinality and optionality. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD. (15 marks)
- (iii) Based on the answer in Q4(b)(i), identify **TWO (2)** associate entities. (2 marks)
- (iii) Identify **ONE (1)** strong relationship and **ONE (1)** weak relationship. Justify your answer. (4 marks)
- [Total : 25 marks]
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